

Phosphate plant -- Diagnostics

Section report: Diagnostics

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-12-31T23:00:00 (365 days, 8568 operating hours)
Product	Dap from 29.0 % P2O5 ore, 1721.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 8760 hourly intervals of 64 signals generated 2026-09-23T14:45:05.265 basis :screening_plant_model

Fault detection and diagnostics

DIAGNOSTIC SUMMARY

Quantity	Value
Findings	99 (29 critical, 65 warning, 5 info)
Value at stake	1.68577175e8 :USD a year
Hours in alarm	19,451 h
Rules applied	20 rules of the catalogue
Deviations caught	9 of 15 (60 %)
Status	:action_required

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	27.63	29.5	:wt_pct	6.33	2026-12-06T10:00:00	2026-12-11T03:00:00	114	1.7546367e7
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	28.59	29.5	:wt_pct	3.09	2026-03-13T10:00:00	2026-03-21T00:00:00	183	8.557122e6
Critical	:evaporation	:acid_strength_low	:STRONG_ACID_P2O5	48.64	50.5	:wt_pct	3.68	2026-12-06T12:00:00	2026-12-11T03:00:00	112	3.839086e6
Critical	:evaporation	:acid_strength_low	:STRONG_ACID_P2O5	48.74	50.5	:wt_pct	3.49	2026-03-13T12:00:00	2026-03-16T03:00:00	64	3.639941e6
Critical	:attack	:sulfate_deficit	:FREE_SO4	0.7732	1.2	:wt_pct	35.56	2026-12-06T10:00:00	2026-12-11T00:00:00	111	3.208715e6
Critical	:attack	:sulfate_deficit	:FREE_SO4	0.7966	1.2	:wt_pct	33.62	2026-03-13T09:00:00	2026-03-16T01:00:00	65	3.033186e6
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.27	5	:count	5.45	2026-07-01T15:00:00	2026-07-01T23:00:00	9	522,738
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.26	5	:count	5.25	2026-06-29T12:00:00	2026-06-30T07:00:00	20	504,030
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.25	5	:count	5.06	2026-06-17T12:00:00	2026-06-18T17:00:00	30	486,215
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.2	5	:count	4	2026-06-22T18:00:00	2026-06-23T04:00:00	11	384,284
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.19	5	:count	3.88	2026-06-25T18:00:00	2026-06-26T02:00:00	9	372,409
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.19	5	:count	3.82	2026-06-24T05:00:00	2026-06-24T23:00:00	19	366,933
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.18	5	:count	3.56	2026-06-27T18:00:00	2026-06-28T21:00:00	28	342,231
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.17	5	:count	3.4	2026-06-20T00:00:00	2026-06-21T05:00:00	30	326,546
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.16	5	:count	3.26	2026-06-16T05:00:00	2026-06-16T14:00:00	10	313,342
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.16	5	:count	3.15	2026-06-16T18:00:00	2026-06-17T10:00:00	17	302,638
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.12	5	:count	2.46	2026-06-19T08:00:00	2026-06-19T13:00:00	6	235,793
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.12	5	:count	2.31	2026-06-22T07:00:00	2026-06-22T09:00:00	3	221,658
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.11	5	:count	2.17	2026-06-25T01:00:00	2026-06-25T14:00:00	14	208,492
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.1	5	:count	2.07	2026-06-26T10:00:00	2026-06-26T15:00:00	6	198,273
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.1	5	:count	2.03	2026-06-30T16:00:00	2026-06-30T21:00:00	6	194,981
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.1	5	:count	1.97	2026-06-23T11:00:00	2026-06-23T14:00:00	4	189,305
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.09	5	:count	1.8	2026-06-23T19:00:00	2026-06-24T00:00:00	6	172,611
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.08	5	:count	1.57	2026-07-01T02:00:00	2026-07-01T04:00:00	3	150,423
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.34	17.4	:wt_pct	6.08	2026-12-06T08:00:00	2026-12-11T06:00:00	119	145,730
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.4	17.4	:wt_pct	5.77	2026-03-13T08:00:00	2026-03-16T06:00:00	71	138,139
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.06	5	:count	1.29	2026-06-26T23:00:00	2026-06-27T01:00:00	3	123,642

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.05	5	:count	1.07	2026-06-21T19:00:00	2026-06-21T22:00:00	4	102,354
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.96	17.4	:wt_pct	2.55	2026-08-09T06:00:00	2026-08-20T03:00:00	262	61,141
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.32	4	:t_per_m2_d	16.92	2026-12-06T16:00:00	2026-12-11T00:00:00	105	4.1237831e7
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.35	4	:t_per_m2_d	16.21	2026-03-13T16:00:00	2026-03-16T00:00:00	57	3.9506143e7
Warning	:beneficiation	:grind_coarse	:CYCLONE_P80	191.39	185	:um	3.45	2026-03-16T21:00:00	2026-03-21T02:00:00	102	9.006969e6
Warning	:attack	:reactor_temperature_high	:ATTACK_TEMP	83.53	83	:deg_c	0.6395	2026-02-09T22:00:00	2026-02-10T19:00:00	22	1.247021e6
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.74	5.6	:count	15.38	2026-12-07T10:00:00	2026-12-11T01:00:00	88	1.240355e6
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.8	5.6	:count	14.32	2026-03-13T04:00:00	2026-03-16T01:00:00	70	1.155143e6
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.87	5.6	:count	13.04	2026-12-06T04:00:00	2026-12-07T05:00:00	26	1.051437e6
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,431.75	4,600	:kw	18.08	2026-12-28T01:00:00	2026-12-31T23:00:00	95	565,587
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,415.2	4,600	:kw	17.72	2026-11-30T01:00:00	2026-12-06T11:00:00	155	554,335
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,412.39	4,600	:kw	17.66	2026-04-20T01:00:00	2026-04-26T20:00:00	164	552,428
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,412.23	4,600	:kw	17.66	2026-01-12T02:00:00	2026-01-18T21:00:00	164	552,317
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,411.35	4,600	:kw	17.64	2026-05-04T02:00:00	2026-05-10T20:00:00	163	551,719
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,408.04	4,600	:kw	17.57	2026-10-05T01:00:00	2026-10-11T21:00:00	165	549,466
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,406.06	4,600	:kw	17.52	2026-09-07T01:00:00	2026-09-13T21:00:00	165	548,119
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,405.86	4,600	:kw	17.52	2026-05-25T01:00:00	2026-05-31T21:00:00	165	547,981
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,404.97	4,600	:kw	17.5	2026-09-28T01:00:00	2026-10-04T21:00:00	165	547,377
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,404.83	4,600	:kw	17.5	2026-03-30T01:00:00	2026-04-05T20:00:00	164	547,284
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,403.99	4,600	:kw	17.48	2026-10-26T01:00:00	2026-11-01T21:00:00	165	546,714
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,403.21	4,600	:kw	17.46	2026-01-19T01:00:00	2026-01-25T20:00:00	164	546,180
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.77	4,600	:kw	17.41	2026-04-13T01:00:00	2026-04-19T21:00:00	165	544,524
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.38	4,600	:kw	17.4	2026-05-11T02:00:00	2026-05-17T21:00:00	164	544,256
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,399.98	4,600	:kw	17.39	2026-01-05T01:00:00	2026-01-11T20:00:00	164	543,984
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,399.75	4,600	:kw	17.39	2026-02-09T01:00:00	2026-02-15T21:00:00	165	543,833
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,399.57	4,600	:kw	17.38	2026-03-23T01:00:00	2026-03-29T21:00:00	165	543,707
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.8	4,600	:kw	17.37	2026-11-02T01:00:00	2026-11-08T20:00:00	164	543,184
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.64	4,600	:kw	17.36	2026-01-01T05:00:00	2026-01-04T21:00:00	89	543,078
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.1	4,600	:kw	17.35	2026-12-14T01:00:00	2026-12-20T22:00:00	166	542,708
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.8	4,600	:kw	17.34	2026-04-27T02:00:00	2026-05-03T21:00:00	164	542,503
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.4	4,600	:kw	17.33	2026-04-06T02:00:00	2026-04-12T21:00:00	164	542,230
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,396.87	4,600	:kw	17.32	2026-10-19T01:00:00	2026-10-25T22:00:00	166	541,869
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,395.82	4,600	:kw	17.3	2026-03-02T01:00:00	2026-03-08T21:00:00	165	541,160
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,395.46	4,600	:kw	17.29	2026-09-14T02:00:00	2026-09-20T20:00:00	163	540,911
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,395	4,600	:kw	17.28	2026-02-16T02:00:00	2026-02-22T20:00:00	163	540,597

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,393.11	4,600	:kw	17.24	2026-01-26T01:00:00	2026-02-01T21:00:00	165	539,312
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,392.84	4,600	:kw	17.24	2026-11-23T01:00:00	2026-11-29T21:00:00	165	539,133
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,392.78	4,600	:kw	17.23	2026-11-16T01:00:00	2026-11-22T21:00:00	165	539,087
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,392.06	4,600	:kw	17.22	2026-10-12T01:00:00	2026-10-18T21:00:00	165	538,601
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,391.57	4,600	:kw	17.21	2026-12-21T01:00:00	2026-12-27T21:00:00	165	538,268
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,390.96	4,600	:kw	17.19	2026-02-23T02:00:00	2026-03-01T21:00:00	164	537,854
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,388.38	4,600	:kw	17.14	2026-11-09T01:00:00	2026-11-15T20:00:00	164	536,100
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,388.33	4,600	:kw	17.14	2026-05-18T01:00:00	2026-05-24T21:00:00	165	536,063
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,385.06	4,600	:kw	17.07	2026-09-21T01:00:00	2026-09-27T21:00:00	165	533,842
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,384.02	4,600	:kw	17.04	2026-02-02T01:00:00	2026-02-08T21:00:00	165	533,134
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,381.81	4,600	:kw	17	2026-03-09T01:00:00	2026-03-13T11:00:00	107	531,632
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,380.66	4,600	:kw	16.97	2026-08-31T06:00:00	2026-09-06T20:00:00	159	530,850
Warning	:beneficiation	:slurry_density_high	:SLURRY_DENSITY	44.37	44	:wt_pct	0.8496	2026-10-06T10:00:00	2026-10-16T00:00:00	231	527,013
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,343.2	4,600	:kw	16.16	2026-07-06T02:00:00	2026-07-19T15:00:00	326	505,375
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,269.36	4,600	:kw	14.55	2026-12-11T13:00:00	2026-12-13T20:00:00	56	455,163
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,248.14	4,600	:kw	14.09	2026-06-29T05:00:00	2026-07-05T19:00:00	159	440,738
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,232.59	4,600	:kw	13.75	2026-07-20T06:00:00	2026-07-26T15:00:00	154	430,161
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,227.32	4,600	:kw	13.64	2026-08-10T05:00:00	2026-08-16T16:00:00	156	426,575
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,225.65	4,600	:kw	13.6	2026-07-27T06:00:00	2026-08-02T16:00:00	155	425,442
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,224.1	4,600	:kw	13.57	2026-06-15T06:00:00	2026-06-21T16:00:00	155	424,386
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,224.06	4,600	:kw	13.57	2026-06-22T05:00:00	2026-06-28T16:00:00	156	424,363
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,220.07	4,600	:kw	13.48	2026-06-01T02:00:00	2026-06-07T15:00:00	158	421,646
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,218.28	4,600	:kw	13.44	2026-06-08T06:00:00	2026-06-14T15:00:00	154	420,434
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,217.21	4,600	:kw	13.42	2026-08-17T06:00:00	2026-08-23T16:00:00	155	419,706
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,215.89	4,600	:kw	13.39	2026-08-24T06:00:00	2026-08-30T15:00:00	154	418,806
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,211.5	4,600	:kw	13.29	2026-08-03T06:00:00	2026-08-09T16:00:00	155	415,817
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	4,911.93	4,600	:kw	6.78	2026-03-16T20:00:00	2026-03-22T21:00:00	146	212,115
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.29	19.2	:wt_pct	0.4718	2026-10-31T03:00:00	2026-11-02T00:00:00	46	181,809
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.27	19.2	:wt_pct	0.3742	2026-10-25T11:00:00	2026-10-26T10:00:00	24	144,210
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.26	19.2	:wt_pct	0.3298	2026-10-26T12:00:00	2026-10-29T06:00:00	67	127,100
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.24	19.2	:wt_pct	0.2078	2026-10-29T16:00:00	2026-10-30T15:00:00	24	80,073
Warning	:utilities	:pump_vibration_high	:PUMP_VIBRATION	6.81	6	:count	13.48	2026-07-21T04:00:00	2026-07-31T00:00:00	237	1,011.27
Info	:filtration	:vacuum_low	:FILTER_VACUUM	163.84	200	:mmwc	18.08	2026-12-06T06:00:00	2026-12-11T01:00:00	116	173,571
Info	:filtration	:vacuum_low	:FILTER_VACUUM	165.13	200	:mmwc	17.43	2026-03-13T06:00:00	2026-03-16T01:00:00	68	167,354
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	546.05	336	:l_ph	62.51	2026-01-01T05:00:00	2026-03-13T08:00:00	1,708	121,827

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	543.42	336	:l_ph	61.73	2026-12-11T13:00:00	2026-12-31T23:00:00	491	120,302
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	537.77	336	:l_ph	60.05	2026-03-16T13:00:00	2026-12-06T08:00:00	6,356	117,028

DID THE DETECTOR CATCH WHAT THE CAMPAIGN INJECTED?

Injected deviation	Role	Kind	Magnitude	Unit	Hours	Detected	Rule that caught it	Findings
:cooling_fouling	:reactor_temperature	:drift	6.2	:deg_c	264	1	:reactor_temperature_high	1
:sulfate_deficit	:sulfate_ratio	:bias	-1.7	:wt_pct	288	0	:none	0
:grind_coarse	:grind_p80	:bias	42	:um	192	1	:grind_coarse	1
:reagent_starvation	:concentrate_grade	:bias	-1.5	:wt_pct	264	1	:reagent_demand_high	1
:filter_cloth_blinding	:filtration_rate	:drift	-1.9	:t_per_m2_d	168	0	:none	0
:evaporator_fouling	:steam_flow	:drift	13.5	:t_ph	360	0	:none	0
:acid_strength_loss	:acid_strength	:drift	-1.6	:wt_pct	312	0	:none	0
:scrubber_breakthrough	:stack_fluoride	:bias	2.4	:count	384	1	:stack_fluoride_high	20
:mill_liner_wear	:mill_load	:drift	430	:kw	264	1	:mill_power_excess	2
:pump_cavitation	:pump_vibration	:bias	3.6	:count	240	1	:pump_vibration_high	1
:sparger_blockage	:product_n	:bias	-1.2	:wt_pct	264	1	:product_nitrogen_low	1
:crusher_gap_drift	:oversize_pct	:bias	6.5	:pct	264	0	:none	0
:rock_grade_drift	:rock_grade	:drift	1.3	:wt_pct	192	0	:none	0
:cyclone_roping	:slurry_density	:bias	6.4	:wt_pct	240	1	:slurry_density_high	1
:ammonia_overfeed	:product_n	:bias	1.1	:wt_pct	192	1	:product_nitrogen_high	4

The catalogue of the detector holds 20 rules, each with the signal it watches, the reference it compares against and the window it averages over; 'PHOSPHATE.RULE_CATALOGUE' prints them and the notebook shows them rule by rule.